

A Gravitational Scaling Law for Atrial Re-Entrant Vulnerability: A Dimensionless Predictor of the Critical Gravity for Spaceflight Atrial Fibrillation

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Abstract

Background: Long-duration spaceflight remodels the atrium: a headward fluid shift dilates the chamber, and atrial electrophysiology changes in ways that raise markers of atrial fibrillation (AF) risk, although sustained AF has not been observed in astronauts. As permanent habitation of the Moon (0.16 g) and Mars (0.38 g) is planned, it is unknown whether a gravitational threshold exists below which the atrium acquires an arrhythmogenic substrate, and where the Moon and Mars lie relative to it. Cardiac physiology has scaling laws with body mass but none with gravitational acceleration.

Methods: Building on the classical wavelength hypothesis of re-entry, we recast the path-length-to-wavelength ratio as a gravity-dependent dimensionless quantity, the Cardiac Gravitational Number, $\mathcal{N}_g = L_0 \cdot \text{dilation}(g) / \text{WL}(g)$, where $\text{WL} = \text{CV} \cdot \text{APD}_{90}$ is the electrical wavelength; re-entry can be hosted when $\mathcal{N}_g \geq 1$, defining a critical gravity g^* with $\mathcal{N}_g(g^*) = 1$. The novelty is the gravity parametrisation, not the underlying criterion. A validated human-atrial ionic model (Courtemanche–Ramirez–Nattel) was driven by a continuous, literature-anchored map from gravitational level to atrial remodelling, and $\mathcal{N}_g$ was computed across 0–1 g. The prediction was independently corroborated in a two-dimensional reaction–diffusion sheet by measuring rotor (phase-singularity) burden, and sensitivity to atrial size and map shape was quantified.

Results: $\mathcal{N}_g$ rose monotonically as gravity fell (0.47 at Earth to 1.29 in free fall), crossing unity at $g^* \approx 0.22$ — between the Moon ($\mathcal{N}_g \approx 1.07$, above threshold) and Mars ($\mathcal{N}_g \approx 0.82$, below threshold in the analytic model only). The full sheet reproduced the trend (rotor burden zero at Earth, rising as gravity fell) but did **not** confirm Mars protection: Mars-level burden overlapped the Moon's. g^* was robust to the assumed map shape (0.13–0.34) but strongly dependent on atrial size (no vulnerability at $L_0 = 6$ cm; Mars vulnerable at $L_0 = 12$ cm).

Conclusions: A simple dimensionless argument predicts a critical gravity for atrial re-entrant vulnerability between the Moon and Mars, corroborated by an independent model. Its strong dependence on atrial size yields a falsifiable, pre-flight-measurable hypothesis: individual atrial dimensions may set off-world arrhythmia risk.

Keywords: atrial fibrillation, microgravity, space medicine, cardiac electrophysiology, re-entry, mechano-electric feedback, in silico model, scaling law

Introduction

Cardiovascular adaptation is a recognised constraint on long-duration human spaceflight, and spaceflight predisposes to cardiac arrhythmias through fluid redistribution, autonomic dysregulation, and radiation exposure [5]. In microgravity a headward fluid shift raises atrial filling and dilates the chamber; deconditioning and altered loading change atrial structure; and autonomic shifts alter atrial refractoriness. On Earth, each of these is independently implicated in the substrate for atrial fibrillation (AF). In astronauts after six-month missions, Khine et al. reported transient left-atrial enlargement ($+12 \pm 18$ mL) and electrophysiological changes consistent with increased AF risk, although no sustained AF was captured; AF prevalence in active astronauts ($\sim 5\%$) resembles the general population but occurs at a younger age [1].

As humanity plans sustained presence on the Moon (0.16 g) and Mars (0.38 g), a question follows that current evidence does not answer: is there a gravitational threshold below which the atrium acquires an arrhythmogenic substrate, and where do the Moon and Mars fall relative to it? Existing models treat spaceflight remodelling as a binary contrast (1 g versus “microgravity”) and are descriptive. Physiology offers powerful allometric scaling laws with body mass, but none with gravitational acceleration.

We treat gravity as a continuous variable and compress the question into a single dimensionless number, in the spirit of the governing numbers of fluid mechanics. We then test whether an independent, spatially resolved reaction–diffusion model agrees, and quantify how far the prediction can be trusted given its assumptions. This is a hypothesis-generating study: it proposes and tests a scaling argument in silico and does not claim experimental proof.

Materials And Methods

The mechanistic chain. Each link rests on established physiology: (i) on Earth a vertical hydrostatic column $\Delta P = \rho gh$ exists and vanishes as $g \rightarrow 0$, driving a cephalad fluid shift; (ii) increased preload raises atrial volume and, by the law of Laplace ($\sigma = P \cdot r / 2t$), wall stress and myocyte stretch — clinically, left-atrial enlargement in spaceflight [1]; (iii) atrial stretch shortens the refractory period and/or slows conduction velocity (CV) and increases their dispersion via mechano-electric feedback [2]; (iv) the electrical wavelength $WL = CV \cdot APD_{90}$ shortens until it no longer fits the available path length, and wavefronts fragment into re-entrant rotors.

The Cardiac Gravitational Number. We define $\mathcal{N}_g = L(g)/WL(g) = L_0 \cdot \text{dilation}(g) / (CV \cdot APD_{90}(g) / 1000)$, with L_0 the ground atrial path length (human left-atrial circumference $\approx 8\text{--}12$ cm; nominal 8 cm) and $CV \approx 58$ cm/s. Re-entry can be hosted when $\mathcal{N}_g \geq 1$; the critical gravity g^* solves $\mathcal{N}_g(g) = 1$. *The ratio of path length to wavelength is itself the classical wavelength hypothesis of re-entry: Rensma et al. measured a normal atrial wavelength of 14–18 cm and atrial-fibrillation induction below ≈ 7.8 cm [4]. The contribution here is not this criterion but its parametrisation by gravity and the resulting threshold g .*

Continuous gravity $\rightarrow$ remodelling map. A minimal linear map sends g to atrial remodelling through the fluid-shift drive $(1 - g)$: electrical-remodelling severity, dilation, and fibrosis density each scale linearly with $(1 - g)$, calibrated so $g = 1$ recovers a validated ground baseline and $g = 0$ an established microgravity operating point.

Ionic model. APD₉₀ was measured with the Courtemanche–Ramirez–Nattel (CRN) human-atrial action-potential model [3] at 1 Hz to steady state, with AF-type remodelling (reduced I_{CaL} , I_{to} , I_{Kur} ; increased I_{K1}) scaling with severity. All severities were paced in one vectorised run; the $g = 0$ column reproduced the scalar baseline APD₉₀ = 293.8 ms exactly.

Full-sheet corroboration. A two-dimensional monodomain reaction–diffusion sheet (CRN ionics; spatially correlated fibrosis) was built at each gravity level with all tissue knobs driven by the same map; because both models share that map, this is a consistency check under spatially resolved dynamics rather than an independent test. To keep rotor cores resolvable on a tractable grid the sheet uses reduced longitudinal coupling ($d_{long} = 0.06$), giving a conduction velocity below the 58 cm/s single-cell calibration, so the corroboration is a qualitative direction-of-effect check not matched in CV to the analytic model. A broken wavefront was induced and rotor burden quantified as phase-singularity density via the topological-charge method on a time-delayed-voltage phase field.

Sensitivity. Without re-simulating the ionic model, g^* was recomputed for $L_0 \in \{6, 8, 10, 12\}$ cm and fluid-shift drive exponents $p \in \{0.5, 1, 2\}$ in $(1 - g)^p$. All code and parameters are openly available (see Additional Information).

Results

$\mathcal{N}_g$ rose monotonically as gravity fell (Table 1), crossing unity at $g^* \approx 0.22$. The Moon (0.16 g) lies just inside the vulnerable region ($\mathcal{N}_g \approx 1.07$); Mars (0.38 g) lies just outside it in the analytic model ($\mathcal{N}_g \approx 0.82$) — though the full sheet does not confirm Mars protection (see below); interplanetary transit (0 g) is most vulnerable ($\mathcal{N}_g = 1.29$); Earth is safe ($\mathcal{N}_g = 0.47$). Notably, although the model was not calibrated to wavelength data, its Earth wavelength (17.1 cm) falls within the normal range measured by Rensma et al. (14–18 cm) and its free-fall wavelength (8.1 cm) sits at their AF-induction threshold (≈ 7.8 cm) [4] — an unprompted, cross-species consistency check.

Table 1: Cardiac Gravitational Number across gravitational level ($L_0 = 8$ cm, $CV = 58$ cm/s).

g (Earth-g)	Body	APD ₉₀ (ms)	WL (cm)	$\mathcal{N}_g$
0.00	Interplanetary transit	139.3	8.08	1.29
0.16	Moon	≈ 163	≈ 9.4	≈ 1.07
0.38	Mars	≈ 202	≈ 11.7	≈ 0.82
0.50	—	219.8	12.75	0.72
1.00	Earth	294.1	17.06	0.47

The full reaction–diffusion sheet (6-seed ensemble per gravity level, bootstrap 95% CI) reproduced the direction of the law (Table 1b): rotor burden was comparably high across transit, the Moon and Mars ($\approx 9\text{--}12 \times 10^{-4}$), fell sharply between 0.5 and 0.8 g, and was exactly zero at Earth (CI 0–0). The models agree at the extremes and in direction but differ in where the transition completes: the analytic threshold is crisp at $g^* \approx 0.22$, while the spatially resolved model is more pessimistic, sustaining rotors through the Mars level and approaching zero only near 0.5–0.6 g. The single-cell argument is thus an optimistic bound on the vulnerable gravity range.

Table 1b: Full-sheet rotor burden (6-seed ensemble, mean and bootstrap 95% CI).

g (Earth-g)	Body	PS density $\times 10^{-4}$ (mean, 95% CI)
0.00	Interplanetary transit	9.05 (8.14–9.81)
0.16	Moon	11.51 (8.87–14.24)
0.38	Mars	10.85 (9.14–12.43)
0.50	—	4.76 (3.30–5.86)
0.80	—	0.21 (0.00–0.62)
1.00	Earth	0.00 (0.00–0.00)

g^* was robust to the assumed fluid-shift map shape ($g^* \in [0.13, 0.34]$ for $p \in \{0.5, 1, 2\}$) but depended strongly on atrial path length: no crossing (no vulnerability at any gravity) for $L_0 = 6$ cm, $g^* = 0.22$ at 8 cm, 0.41 at 10 cm, and 0.58 at 12 cm — at which point Mars itself becomes vulnerable (Table 2).

Table 2: Sensitivity of the critical gravity g^* .

Assumption	Value	g^*
$L_0 = 6$ cm	Small atrium	None (always safe)
$L_0 = 8$ cm	Nominal	0.22
$L_0 = 10$ cm	Enlarged	0.41
$L_0 = 12$ cm	Dilated	0.58 (Mars vulnerable)
Drive exponent $p = 0.5$	Concave	0.34
$p = 1.0$	Linear	0.22
$p = 2.0$	Convex	0.13

Discussion

A one-line dimensionless argument predicts a critical gravity for atrial re-entrant vulnerability between the Moon and Mars, and an independent reaction–diffusion model corroborates the trend. The prediction is consistent with the clinical picture — increased vulnerability without sustained AF [1] — because $\mathcal{N}_g$ indexes the substrate’s proximity to re-entry, not the occurrence of fibrillation.

The most decision-relevant result is the strong dependence on atrial size: because g^* moves from “no one is ever vulnerable” ($L_0 = 6$ cm) to “even Mars is vulnerable” ($L_0 = 12$ cm), the framework predicts that individual atrial dimensions could determine off-world arrhythmia risk — a personalised, pre-flight-measurable hypothesis, since atrial size is routinely imaged.

This study has clear limitations. First and most important, the microgravity electrical remodelling is parameterised as full chronic-AF-type ionic remodelling (I_{CaL} –70%, I_{to}/I_{Kur} –50%, I_{K1} +100%), shortening APD₉₀ from ≈ 294 to ≈ 139 ms at $g = 0$; there is no spaceflight evidence the atrial APD shortens this much (Khine et al. report only modest, transient change and no sustained AF), so this single assumption sets the magnitude of the

entire curve and is likely an overestimate. Second, the gravity $\rightarrow$ remodelling map is shared by both models, so the sheet is a consistency check of the map's consequences, not an independent test of the map; the map's linear form and its acute-stretch mechano-electric sensitivities (used for chronic adaptation) are hypotheses, not data. The corroboration uses a single ionic model in two dimensions at reduced coupling, without autonomic dynamics or three-dimensional geometry, and its confidence intervals do not separate transit, Moon and Mars. $\mathcal{N}_g$ is proposed, not proven; the analytic threshold is an optimistic bound that the full model does not confirm. No human partial-gravity electrophysiology exists to fix the absolute value of g , so the headline $g \approx 0.22$ is conditional on $L_0 \approx 8$ cm and the stated calibration. The framework is nonetheless falsifiable: partial-gravity measurements of atrial APD₉₀, CV and effective path length that yield an $\mathcal{N}_g$ not crossing unity within $[0, 1]$ g , or crossing it far from the predicted band, would refute the hypothesis.

Conclusions

A dimensionless Cardiac Gravitational Number predicts a critical gravity for atrial re-entrant vulnerability that falls between the Moon and Mars and is corroborated by an independent reaction–diffusion model. The prediction depends primarily on atrial size, giving a concrete, falsifiable, pre-flight-measurable hypothesis: that individual atrial dimensions may set off-world arrhythmia risk. The decisive test is partial-gravity atrial electrophysiology, ideally stratified by atrial size.

Additional Information

Author Contributions: LPR conceived the hypothesis, designed and implemented the model and analyses, produced the figures, and wrote the manuscript.

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Data Availability: All code, parameters, and scripts that reproduce every figure and number are openly available at <https://github.com/laurapiro17/atrial-fib-in-microgravity>.

References

1. Khine HW, Steding-Ehrenborg K, Hastings JL, et al.: Effects of prolonged spaceflight on atrial size, atrial electrophysiology, and risk of atrial fibrillation. *Circ Arrhythm Electrophysiol.* 2018, 11:e005959. 10.1161/CIRCEP.117.005959
2. Ravelli F: Mechano-electric feedback and atrial fibrillation. *Prog Biophys Mol Biol.* 2003, 82:137-149. 10.1016/S0079-6107(03)00011-7
3. Courtemanche M, Ramirez RJ, Nattel S: Ionic mechanisms underlying human atrial action potential properties: insights from a mathematical model. *Am J Physiol.* 1998,

275:H301-H321. 10.1152/ajpheart.1998.275.1.H301

4. Rensma PL, Allessie MA, Lammers WJ, Bonke FI, Schalij MJ: Length of excitation wave and susceptibility to reentrant atrial arrhythmias in normal conscious dogs. *Circ Res.* 1988, 62:395-410. 10.1161/01.res.62.2.395
5. Mircea AA, Pistritu DV, Fortner A, et al.: Space travel: the radiation and microgravity effects on the cardiovascular system. *Int J Mol Sci.* 2024, 25:11812. 10.3390/ijms252111812